DEMONSTRATION ON OPEN DATA · NOVEMBER 2026

Site screening memo: Bulgaria

Solar parks and wind farms against protected areas, critical habitat, ecosystem integrity and water risk, in the order of the TNFD Locate phase

SCOPE	Solar parks and wind farms of 1 ha or more mapped in OpenStreetMap, Bulgaria, data of 22 October 2026
FRAMEWORKS	TNFD LEAP, Locate phase · ESRS E4 and GRI 101-5 (sites in or near sensitive areas) · IFC PS6 and EBRD PR6 (critical habitat screening)
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Status. A demonstration on open data, not client work, and not reviewed by a biodiversity specialist. Screening, not assessment: a flag means look closer, never a finding about a site.

Summary

- **504 sites** mapped in OpenStreetMap on 22 October 2026: 392 solar parks (4,415 ha) and 112 wind farms (482 turbines).
- **The headline metric:** 238 sites (47%) are in or within 1 km of a Natura 2000 site or likely critical habitat, the number ESRS E4, GRI 101-5 and TNFD ask for.
- **For lenders:** 121 sites are inside Natura 2000 or on likely critical habitat, where a critical habitat assessment under IFC PS6 or EBRD PR6 is likely to be asked for. Another 117 are within 1 km, worth a check.
- **Wind and birds:** 77 of the 112 wind farms are within 5 km of a Birds Directive site.
- **Water:** 101 sites (20%) are in catchments with high or extremely high physical water risk.
- **Priority locations:** 156 sites to look at first, listed in the download with the reasons.

1 Asset inventory

Solar parks and wind farms mapped as power plants in OpenStreetMap, and the wind turbines outside them, grouped into a farm when they stand within 800 m of each other (the distance used by Dunnett et al. 2020). A farm's footprint is the outline of its turbines plus 150 m: wider than the ground the turbines stand on, narrower than a leased area. Sites under 1 ha are left out. With a client, the client's own site register replaces this list.

TYPE	SITES	AREA (HA)	MEDIAN (HA)	OFFICIAL CAPACITY, END OF 2024 (EUROSTAT)
Solar parks	392	4,415	4.4	4,560 MW, rooftops included
Wind farms	112	8,372 (footprints)	21.7	706 MW; OpenStreetMap's turbines add up to about 930 MW

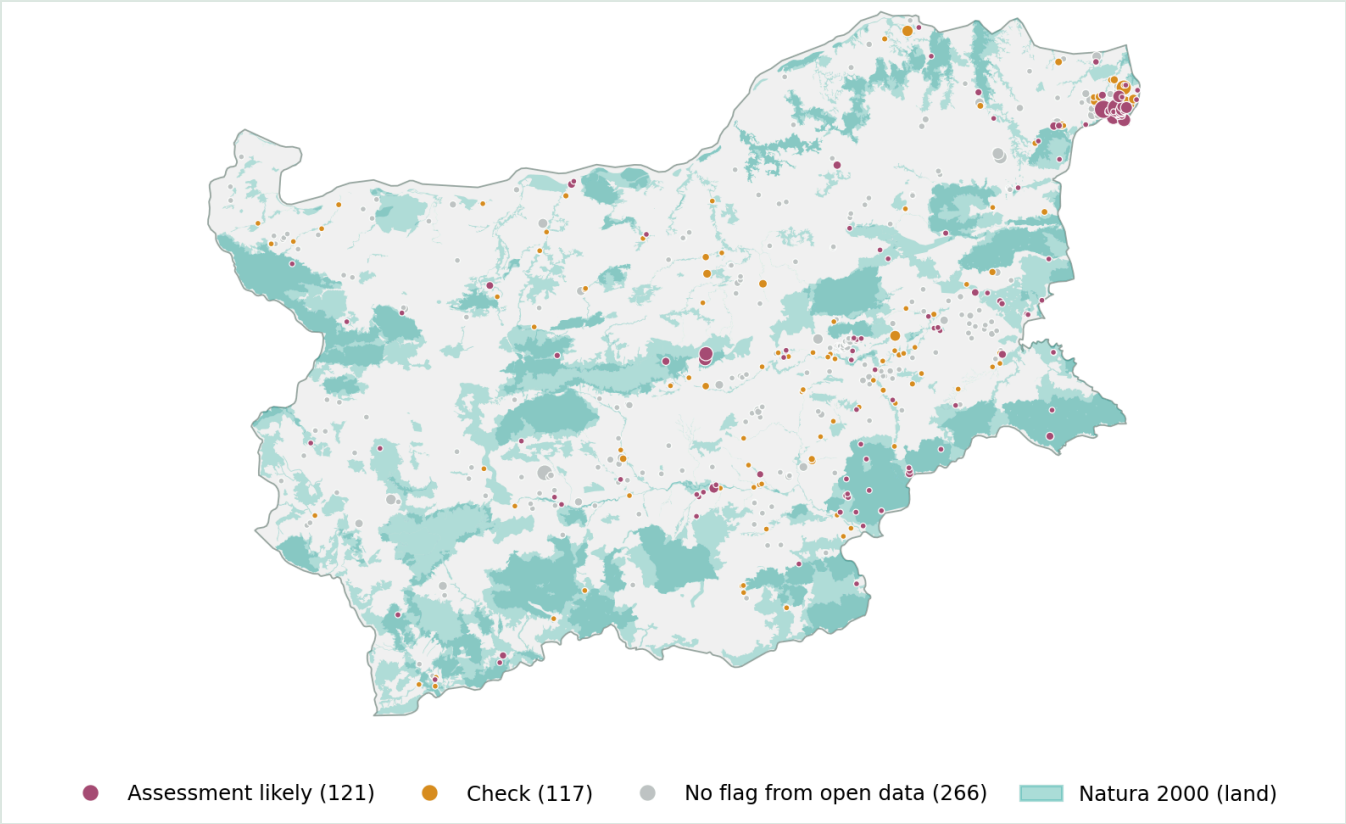


Figure 1. The sites by screening result, with Natura 2000 on land. Dot size: site area. Outlines: Natural Earth. Screening rules in section 5.

2 Interface with nature

What the sites sit on and next to: land cover from ESA WorldCover 2021 on each footprint and within 1 km of it, and the Biodiversity Intactness Index (Impact Observatory and Vizzuality), which estimates how much of the original biodiversity remains, from 0 to 1.

MEASURE	SOLAR PARKS	WIND FARMS
Footprint: cropland	23%	71%
Footprint: grassland	54%	18%
Footprint: trees and shrubs	5%	10%
Footprint: built-up	17%	1%
Within 1 km: cropland	47%	65%
Within 1 km: grassland	24%	13%
Within 1 km: trees and shrubs	19%	19%
Biodiversity intactness within 1 km, 2020 (median)	0.54	0.53

WorldCover maps much of a solar park's footprint as grassland or built-up. For parks built after 2021, it shows the land before the park.

Year by year, within 1 km of the sites

From 2017 to 2025, 0.3% of the land within 1 km of the sites lost tree cover (Hansen/UMD), against 0.7% of 400 random 2 km squares of the country. Natural cover that lasted as crops or built-up land, or came back, netted -0.09% of the land from 2019 to 2022, against +0.14% in the countryside (Impact Observatory annual land cover). 26 sites lost natural or tree cover over at least 5% of the land around them; the figures for each site are in the download.

3 Sensitive locations

TNFD names five kinds of sensitive location. Open data covers four of them here; the thresholds are this screening's choices, written down before the results.

CRITERION	OPEN DATA AND THRESHOLD	SOLAR	WIND	ALL
Important for biodiversity	Natura 2000 (end-2024 release) or likely critical habitat, inside or within 1 km	167	71	238
High ecosystem integrity	Biodiversity Intactness within 1 km in the country's top quarter, 2020	6	8	14
Rapid decline in integrity	A fall within 1 km, 2017 to 2020, steeper than on 95% of the country's land	36	2	38
High physical water risk	Aqueduct 4.0 physical risk, quantity or quality, high or extremely high	100	1	101
Ecosystem services for local communities	Not covered by open data			
Any of the above		243	74	**317**

The headline metric

Sites in or near protected areas or areas of high biodiversity importance (ESRS E4, GRI 101-5, TNFD, and the WBCSD metric): **238 of 504 (47%)**, with near set at 1 km. TNFD doesn't fix the distance and GRI 101-5 asks for it to be stated, so here is how the share near Natura 2000 moves with it:

NATURA 2000	INSIDE	WITHIN 500 M	WITHIN 1 KM	WITHIN 2 KM	WITHIN 5 KM
Sites	101 (20%)	180 (36%)	238 (47%)	318 (63%)	453 (90%)

Of the 121 sites flagged "Assessment likely", Natura 2000 alone flags 45, likely critical habitat alone 23, and both 53. Baseline water stress, which ESRS E3 and GRI 303 use, is high or extremely high for 101 sites.

4 Priority locations

Where to look first: the sites flagged "Assessment likely" (inside Natura 2000 or on likely critical habitat) and the wind farms within 5 km of a Birds Directive site: 156 sites in all. The eight largest flagged for assessment are below; the full list, with every reason, is in the download.

SITE	TYPE (TURBINES)	AREA (HA)	NATURA 2000 WITHIN 1 KM	BIRDS SITE	CRITICAL HABITAT	LOCATION
BG-W0471	wind (30)	1,087	Kaliakra (SPA)	inside	likely	43.46 N, 28.42 E
BG-W0462	wind (28)	1,077	Kaliakra (SPA)	inside	likely	43.47 N, 28.33 E
BG-W0402	wind (14)	393	Bulgarka (SPA and SCI); and 2 more	inside	likely	42.75 N, 25.44 E
BG-W0451	wind (24)	388	Kompleks Kaliakra (SCI); and 1 more	inside	likely	43.42 N, 28.39 E
BG-W0452	wind (35)	315	Kompleks Kaliakra (SCI); and 1 more	inside	likely	43.39 N, 28.45 E
BG-W0401	wind (15)	311	Bulgarka (SPA and SCI); and 2 more	inside	likely	42.72 N, 25.43 E
BG-W0488	wind (8)	239	Kraymorska Dobrudzha (SCI)	2.7 km	potential	43.51 N, 28.45 E
BG-W0480	wind (7)	217	Kaliakra (SPA)	inside	likely	43.44 N, 28.45 E

5 Screening note for IFC PS6 and EBRD PR6

- **Assessment likely (121 sites):** inside a Natura 2000 site or on likely critical habitat. A lender is likely to ask for a critical habitat assessment, and Natura 2000 sites, as legally protected areas, bring extra requirements under both standards.
- **Check (117):** within 1 km of either. A biodiversity specialist should look at the distance and at what the nearby site protects.
- **No flag from open data (266):** nothing found in these layers. That doesn't clear a site, since key biodiversity areas and species records aren't in open data.
- **Potential critical habitat** (397 sites on it) is reported in the download but not flagged, because it covers most of the country's lowlands.
- **Wind farms within 5 km of a Birds Directive site (77):** collision and displacement risk for birds is a question for the assessment, including the cumulative effect of neighbouring farms.

6 What this screening can't tell

- **Key biodiversity areas and species records,** behind licences that don't allow this use. The critical habitat layer stands in for them, at about 1 km resolution.
- **Whether the species that would make a site critical habitat are there,** which needs species data and field survey.
- **Ecosystem services for Indigenous Peoples and local communities,** the fifth TNFD criterion.
- **Exact site boundaries:** OpenStreetMap shows what has been mapped, and grouping turbines can join or split wind farms.
- **Recent change:** Natura 2000 end 2024, Biodiversity Intactness up to 2020, WorldCover 2021.

7 Data and licences

OpenStreetMap, data of 22 October 2026 (© OpenStreetMap contributors, ODbL). Natura 2000, end-2024 release (EEA and DG ENV, CC BY 4.0). Critical Habitat screening layer, basic (UNEP-WCMC and The Biodiversity Consultancy, 2025, CC BY 4.0). Aqueduct 4.0 (WRI, CC BY 4.0). Biodiversity Intactness Index (Impact Observatory and Vizzuality, CC BY 4.0). ESA WorldCover 2021 (CC BY 4.0). Global Forest Change 2025 v1.13 (Hansen/UMD, CC BY 4.0). Drivers of tree cover loss v1.13 (WRI and Google DeepMind, CC BY 4.0). Annual land use land cover v2 (Impact Observatory, CC BY 4.0). Eurostat nrg_inf_epcrw (CC BY 4.0). Natural Earth (public domain). Not used: WDPA and key biodiversity area data, whose terms don't allow commercial use or redistribution without permission, and the non-commercial drill-down version of the critical habitat layer.

8 How AI was used

An agent (Claude Code) wrote the scripts: the site list, the downloads, the screening, the checks, the maps and this memo. The rules were written down before the results, and the results were checked against a known case (the wind farms at Kaliakra, Bulgaria, subject of EU Court of Justice case C-141/14), the official capacity statistics and the maps. What it got wrong first: turbines grouped within 1.5 km, which joined neighbouring wind farms; potential critical habitat flagged as if it were likely; and Aqueduct's overall water risk, which counts regulatory and reputational risk, used where TNFD asks about physical risk.

9 Sources

- TNFD (2023). Guidance on the identification and assessment of nature-related issues: the LEAP approach. WBCSD metric: sites in or near protected or key biodiversity areas.
- IFC (2012). Performance Standard 6; Guidance Note 6 (2019). EBRD (2019). Performance Requirement 6 and guidance note.
- Dunnett, S. et al. (2025). An update to the global Critical Habitat screening layer. Scientific Data 12, 1812. Dunnett, S. et al. (2020). Harmonised global datasets of wind and solar farm locations and power. Scientific Data 7, 130.
- Kuzma, S. et al. (2023). Aqueduct 4.0. World Resources Institute. Zanaga, D. et al. (2022). ESA WorldCover 10 m 2021 v200.